AI TRAINING – WEEK 3
DAY 1 ASSIGNMENT

Pandas Basics & Exploratory Data Analysis

Load → Explore → Filter → Answer Business Questions

1. What You Will Learn

Pandas helps you work with data arranged in rows and columns. Load a real CSV dataset, explore it, filter records, and answer business questions using evidence from the data.

- Series and DataFrame
- Read a CSV using pd.read_csv()
- Inspect data using head(), info(), and describe()
- Select columns and filter rows
- Identify missing values and common data issues
- Answer business questions using data
- Explain findings in simple language

2. Student Deliverable – Keep It Simple

Submit ONE Jupyter Notebook only.

File name: Week3_Day1_EDA_YourName.ipynb

Include code, outputs, answers to all 15 questions, and short explanations. Keep the CSV in your project folder or include its source link. No screenshots or separate Word/PDF report are required.

3. Choose a Dataset

Choose a real, publicly accessible CSV with enough rows and useful columns to answer at least 15 questions.

- Retail / Superstore sales – sales, profit, category, region, date
- Online store orders – products, quantity, price, customer, order date
- Food delivery orders – restaurant, order value, city, delivery time
- Student performance – scores, attendance, study time

Recommended: retail or Superstore sales. Write the dataset name and source URL at the top of your notebook.

4. Part A – Load and Inspect the Dataset

```
import pandas as pd

df = pd.read_csv("your_dataset.csv")
```

- Display the first five rows using head().

```
import pandas as pd
df=pd.read_csv("students.CSV")
print(df.head())
```

- Display the last five rows using tail().

```
import pandas as pd
df=pd.read_csv("students.CSV")
print(df.tail(5))
```

- Use info() to inspect columns and data types.

```
import pandas as pd
df=pd.read_csv("students.CSV")
print(df.info())
```

output:

```
#   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   name    5 non-null      str
 1   age     6 non-null      str
 2   marks   6 non-null      int64
 3   city    5 non-null      str
dtypes: int64(1), str(3)
memory usage: 324.0 bytes
None
```

- Use describe() to summarize numeric columns.

```
import pandas as pd
df=pd.read_csv("students.CSV")
print(df.describe())
```

output:

```
            marks
count    6.000000
mean    85.833333
std      7.467708
min     75.000000
25%     81.250000
50%     86.500000
75%     91.000000
max     95.000000
```

- Print the number of rows and columns using shape.

```
import pandas as pd
df=pd.read_csv("students.CSV")
print(df.shape)
```

output:
(6, 4)

- Display all column names.

```
import pandas as pd
df=pd.read_csv("students.CSV")
print(df.columns)
```

output: Index(['name', 'age', 'marks', 'city'], dtype='str')

- Identify columns containing text, numbers, dates, or categories

```
import pandas as pd
df=pd.read_csv("students.CSV")
for column in df.columns:
    print(column, ":", df[column].dtype)
```

output:
name : str
age : str
marks : int64
city : str.

5. Part B – Series and DataFrame

- Select one column and store it in a variable. Explain why it is a Series.

```
import pandas as pd
df=pd.read_csv("students.CSV")
marks=df["marks"]
print(marks)
print(type(marks))
```

ans:it is one dimensional having only one coloumn.

- Select two or more columns and store them in another variable. Explain why this is a DataFrame.

```
import pandas as pd
df=pd.read_csv("students.CSV")
details=df[["name","marks"]]
print(details)
print(type(details))
```

output:

	name	marks
0	Rahul	85
1	Priya	92
2	NaN	75
3	Arjun	80
4	Sneha	95
5	Kiran	88

<class 'pandas.DataFrame'>

- Print type() for each variable.

```
import pandas as pd
df=pd.read_csv("students.CSV")
marks=df["marks"]
details=df[["name","marks"]]
print(type(marks))
print(type(details))
```

- output: <class 'pandas.Series'>
  <class 'pandas.DataFrame'>
- Explain the difference between a Series and a DataFrame in your own words.

  Ans: a dataset is said to be series if it is having only one coloums (1 dimensional).dataset is said to be dataframe if it is having one or more column. Two dimensional data structure.

6. Part C – Filtering Practice

Adapt the examples to column names in your chosen dataset.

high_value = df[df["Sales"] > 500]
selected = df[df["Region"] == "West"]
combined = df[(df["Sales"] > 500) & (df["Profit"] > 0)]

- Filter rows using one condition.

```
import pandas as pd
df=pd.read_csv("students.CSV")
high_marks=df[df["marks"] >90]
print(high_marks)
```

- Filter rows using two conditions with & or |.

```
import pandas as pd
df=pd.read_csv("students.CSV")
high_marks=(df["marks"] > 90) & (df["city"].isna())
print(high_marks)
```

- Filter rows using a text/category value.

```
import pandas as pd
df=pd.read_csv("students.CSV")
city_students = df[df["city"] == "Hyderabad"]
print(city_students)
```

- Display selected columns from filtered data.

```
import pandas as pd
df=pd.read_csv("students.CSV")
high_marks = df[df["marks"] > 80]
selected_data = high_marks[["name", "marks"]]
print(selected_data)
```

output:

	name	marks
0	Rahul	85
1	Priya	92
4	Sneha	95
5	Kiran	88

- Count how many rows match each filter.

```
import pandas as pd
df=pd.read_csv("students.CSV")
high_marks = df[df["marks"] > 80]
count = len(high_marks)
print("Number of students who scored above 80:", count)
```

- If dates exist, filter a chosen date range.

Important: Put each condition in parentheses when combining conditions with & or |.

7. Part D – Answer 15 Business Questions

For every answer, include the code, result/output, and one sentence explaining what it means. Adapt questions to the columns in your dataset where needed.

output: (6, 4)

1. How many rows and columns are in the dataset?

```
import pandas as pd
df=pd.read_csv("students.CSV")

print(df.shape)
```

output: (6, 4)

Finding: The dataset contains 6 student records and 4 columns

2. What time period does the dataset cover, if dates are available?

```
import pandas as pd
df=pd.read_csv("students.CSV")
date_columns = [col for col in df.columns
                if "date" in col.lower()]

print(date_columns)
```

output: output: (6, 4)

Finding: The dataset has no date columns, so its time period cannot be determined.

3.Which columns are available, and what do the important columns represent?

```
import pandas as pd
df=pd.read_csv("students.CSV")
print(df.columns.tolist())
print(df.dtypes)
```

output: ['name', 'age', 'marks', 'city']

name object
age object
marks int64
city object

Finding: The dataset contains student names, ages, marks, and cities; `marks` is numeric, while `name`, `age`, and `city` are stored as object data because the age column includes a non-numeric value (`abc`)

4. Which product, category, or item appears most often?

```
import pandas as pd
df=pd.read_csv("students.CSV")

print(df["city"].value_counts())
```

city

Hyderabad 2

Chennai 2

Bangalore 1

Finding: Hyderabad and Chennai are tied for the most student records, with 2 students each; the missing city is excluded by default.

5. What is the total sales/revenue, if a sales or amount column exists?

```
import pandas as pd
df=pd.read_csv("students.CSV")

print("Total marks:", df["marks"].sum())
```

Total marks: 515

Finding: The sum of marks obtained by all 6 students is 515.

6. What is the average order value, score, or relevant numeric measure?

```
import pandas as pd
df=pd.read_csv("students.CSV")
```

```
print("Average marks:", round(df["marks"].mean(), 2))
```

Average marks: 85.83

Finding: The average mark across the 6 students is 85.83.

7. Which record has the highest value in the main numeric column?

```
import pandas as pd
df=pd.read_csv("students.CSV")

highest = df.loc[df["marks"].idxmax()]
print(highest)
```

name Sneha

age 21

marks 95

city NaN

Finding: Sneha obtained the highest marks, with 95, and her city value is missing.

8. Which record has the lowest value in the main numeric column?

```
import pandas as pd
df=pd.read_csv("students.CSV")

lowest = df.loc[df["marks"].idxmin()]
print(lowest)
```

name NaN

age 20

marks 75

city Chennai

Finding: The student with the missing name scored the lowest marks, with 75, and is listed in Chennai

9. Which category or group has the highest total value?

```
import pandas as pd
df=pd.read_csv("students.CSV")
print(df.groupby("city")["marks"].sum().sort_values(ascending=False))
```

city

Hyderabad 173

Chennai 155

Bangalore 92

Finding: Hyderabad has the highest total marks, with 173 across its two students; records with missing city values are excluded from this grouping.

10. Which region, city, or location has the most records?

```
import pandas as pd
df=pd.read_csv("students.CSV")
print(df["city"].value_counts())
```

city

Hyderabad 2

Chennai 2

Bangalore 1

Finding: Hyderabad and Chennai tie for the highest number of student records, with 2 each

11. How many records meet a condition you define, such as sales above a threshold?

```
import pandas as pd
df=pd.read_csv("students.CSV")

high_marks = df[df["marks"] > 90]
print(high_marks)
print("Count:", len(high_marks))
```

name age marks city

1 Priya 22 92 Bangalore

4 Sneha 21 95 NaN

Count: 2

Finding: Two students, Priya and Sneha, scored more than 90 marks.

12. What percentage of records meet that condition?

Finding: Approximately 33.33% of the students scored more than 90 marks.

13. Are there missing values? Which columns have the most missing values?

Finding: The name and city columns each have one missing value, while `age` and `marks` have no missing values.

14. Are there duplicate rows? How many?

Finding: The dataset contains no completely duplicated rows.

15. Write three useful observations or business recommendations based on your findings.

Finding:

8. Part E – Data Quality Check

- Check missing values using isna().sum().

```
import pandas as pd
df=pd.read_csv("students.CSV")

print(df.isna().sum())
```

- Check duplicate rows using duplicated().sum().

```
import pandas as pd
df=pd.read_csv("students.CSV")
print("Duplicate rows:", df.duplicated().sum())
```

- Inspect data types and identify columns that should be numeric or dates.

```
import pandas as pd
df=pd.read_csv("students.CSV")
print(df.dtypes)
```

- If dates exist, convert them using pd.to_datetime() where appropriate.

```
import pandas as pd
df=pd.read_csv("students.CSV")
print(df[pd.to_numeric(df["age"], errors="coerce").isna()])
```

- Do not remove missing values or duplicates automatically; first explain what you found.

9. Part F – Common Errors to Investigate

Investigate each issue, write the cause, and show how you corrected it.

- FileNotFoundError – incorrect CSV path or filename.
- KeyError – column name does not match df.columns exactly.
- TypeError – a numeric operation is attempted on text data.
- Incorrect filtering – combined conditions are missing parentheses.
- Unexpected results – missing values or incorrect data types affect calculations.

10. Part G – Explain in Your Own Words

1. What is Pandas used for?

Answer: pandas is a python library which is used for data manuplation.

2. What is a Series?

Answer: which is one dimensional data structure

3. What is a DataFrame?

Answer: contains two dimensional data structurs

4. What do head(), info(), and describe() tell you?

Answer:head() shows the first rows,info() shows data types, and describe() shows statistical summaries

5. What does filtering mean?

Answer: Filtering means selecting rows based on specific conditions.

6. Why should you inspect data before analyzing it?

Answer: Inspecting data helps identify missing values, incorrect data, and duplicates before analysis.

7. What issue did you find in your dataset, and how would you investigate it?

Answer: I found missing values in name and city and an invalid abc value in age which I investigated by checking the affected rows.

8. Which finding from your analysis was most useful?

Answer: Identifying students who scored above 90 marks was the most useful finding.

11. Student Submission Checklist

- I recorded the dataset name and source.
- My CSV loads successfully.
- I used head(), info(), describe(), and shape.
- I demonstrated Series and DataFrame.
- I filtered with one and multiple conditions.
- I answered all 15 business questions using evidence.
- I checked missing values and duplicates.
- I investigated and corrected common errors.
- I explained findings in my own words.
- My notebook runs from top to bottom.

12. Daily Deliverable – EDA Notebook v1

File name: Week3_Day1_EDA_YourName.ipynb

Submit one notebook containing code, outputs, 15 answered business questions, data-quality checks, and brief explanations. Organize it with simple headings.

Key Principle: Don't just run Pandas commands. Use the data to answer a question, show the evidence, and explain what it means.